24 A When the switch is closed, the capacitor charges up until it reaches the e.m.f. of the battery V_0.

P.d. across the capacitor increases with time: $V_c = V_0(1 - e^{-\frac{t}{RC}})$

Hence, the p.d. across the resistor decreases with time : $V_R = V_0 e^{-\frac{t}{RC}}$

[the p.d. across the resistor was initially the e.m.f. of the battery, and the current in the circuit is V_0/R.]

Power dissipated by the resistor, $P = \frac{V_R^{\,2}}{R} = \frac{V_0^{\,2}}{R} e^{-\frac{2t}{RC}}$

Or simply consider that the current varies in the circuit (and therefore through the resistor) as: $I_R = I_0 e^{-\frac{t}{RC}}$ and the power in R is $P = I_R^{\,2} R = I_0^{\,2} R e^{-\frac{2t}{RC}}$

The power in R follows an exponential decay, hence there is a constant "half-life" (just as in radioactive decay).

Option A shows an **exponential decay of power with a constant half-life**.

25 C The magnetic force acting on the proton is upwards. To balance this force, so that the proton